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1. Experiment Title:

Create a Schedule Planner for Given table and shows the Alert Box() in Click Event.

2. Software/Tools Required:

Google Chrome (Web Browser), Visual Studio Code (Code Editor), Browser Developer Tools / Console, HTML, CSS, Javascript

3. Experiment Program Code:

Case Study 9.html

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Schedule Planner</title>
    <link rel="stylesheet" href="Case Study 9.css">
    <link rel="preconnect" href="https://fonts.googleapis.com">
    <link rel="preconnect" href="https://fonts.gstatic.com" crossorigin>
    <link href="https://fonts.googleapis.com/css2?family=Inter:wght@300;400;500;600;700&display=swap"
rel="stylesheet">
</head>
<body>
    <div class="container">
        <h2>Schedule Planner</h2>
        <p class="subtitle">Click on any schedule session or topic to view details.</p>

        <div class="table-wrapper">
            <table class="schedule-table" id="scheduleTable">
                <thead>
                    <tr>
                        <th rowspan="3" class="th-main">Day</th>
                        <th colspan="3" class="th-main">Seminar</th>
                    </tr>
                    <tr>
                        <th colspan="2" class="th-sub">Schedule</th>
                        <th rowspan="2" class="th-main">Topic</th>
                    </tr>
                    <tr>
                        <th class="th-sub">Begin</th>
                        <th class="th-sub">End</th>
                    </tr>
                </thead>
                <tbody>
                    <tr>
                        <td rowspan="2" class="day-cell" data-day="Monday">Monday</td>
                        <td rowspan="2" class="slot-yellow" data-day="Monday" data-time="8:00 a.m. - 5:00
p.m." data-topic="Introduction to XML & Validity: DTD and Relax NG">8:00 a.m.</td>
                        <td rowspan="2" class="slot-purple" data-day="Monday" data-time="8:00 a.m. - 5:00
p.m." data-topic="Introduction to XML & Validity: DTD and Relax NG">5:00 p.m.</td>
                        <td class="topic-cell" data-day="Monday" data-time="8:00 a.m. - 12:00 p.m." data-
topic="Introduction to XML">Introduction to XML</td>
                    </tr>
                    <tr>
```

```
                        <td class="topic-cell" data-day="Monday" data-time="1:00 p.m. - 5:00 p.m." data-
topic="Validity: DTD and Relax NG">Validity: DTD and Relax NG</td>
                    </tr>
                    <tr>
                        <td rowspan="3" class="day-cell" data-day="Tuesday">Tuesday</td>
                        <td class="slot-yellow" data-day="Tuesday" data-time="8:00 a.m. - 11:00 a.m."
data-topic="XPath">8:00 a.m.</td>
                        <td class="slot-yellow" data-day="Tuesday" data-time="8:00 a.m. - 11:00 a.m."
data-topic="XPath">11:00 a.m.</td>
                        <td class="topic-cell" data-day="Tuesday" data-time="8:00 a.m. - 11:00 a.m." data-
 topic="XPath">XPath</td>
                    </tr>
                    <tr>
                        <td class="slot-lightgreen" data-day="Tuesday" data-time="11:00 a.m. - 2:00 p.m."
data-topic="XPath (Continuation)">11:00 a.m.</td>
                        <td class="slot-lightgreen" data-day="Tuesday" data-time="11:00 a.m. - 2:00 p.m."
data-topic="XPath (Continuation)">2:00 p.m.</td>
                        <td rowspan="2" class="topic-cell" data-day="Tuesday" data-time="11:00 a.m. - 5:00
p.m." data-topic="XSL Transformations">XSL Transformations</td>
                    </tr>
                    <tr>
                        <td class="slot-lightgreen" data-day="Tuesday" data-time="2:00 p.m. - 5:00 p.m."
data-topic="XSL Transformations">2:00 p.m.</td>
                        <td class="slot-purple" data-day="Tuesday" data-time="2:00 p.m. - 5:00 p.m." data-
topic="XSL Transformations">5:00 p.m.</td>
                    </tr>
                    <tr>
                        <td class="day-cell" data-day="Wednesday">Wednesday</td>
                        <td class="slot-yellow" data-day="Wednesday" data-time="8:00 a.m. - 12:00 p.m."
data-topic="XSL Formatting Objects">8:00 a.m.</td>
                        <td class="slot-lightgreen" data-day="Wednesday" data-time="8:00 a.m. - 12:00
p.m." data-topic="XSL Formatting Objects">12:00 p.m.</td>
                        <td class="topic-cell" data-day="Wednesday" data-time="8:00 a.m. - 12:00 p.m."
data-topic="XSL Formatting Objects">XSL Formatting Objects</td>
                    </tr>
                </tbody>
            </table>
        </div>

        <div class="edit-panel" id="editPanel">
            <h3 style="margin-top: 0; font-size: 16px;">Edit Selected Topic / Item</h3>
            <p id="selectedInfo" style="font-size: 13px; color: #555; margin-bottom: 10px;">Select a cell
from the table to edit.</p>
            <div style="display: flex; gap: 8px; justify-content: center; flex-wrap: wrap;">
                <input type="text" id="editInput" placeholder="Click any table cell to edit"
style="padding: 8px 12px; font-size: 14px; border: 1px solid #ccc; border-radius: 4px; width: 60%; min-
width: 220px;">
                <button type="button" id="saveBtn" style="padding: 8px 16px; background-color: royalblue;
color: white; border: none; border-radius: 4px; font-weight: 600; cursor: pointer;">Save</button>
                <button type="button" id="resetBtn" style="padding: 8px 14px; background-color: #6c757d;
color: white; border: none; border-radius: 4px; font-weight: 600; cursor: pointer;">Reset Default</button>
            </div>
        </div>

        <footer>
            Made by Rohan Laharwani PRN - 24070521055
        </footer>
    </div>

    <script src="Case Study 9.js"></script>
</body>
</html>
```

Case Study 9.css

```
* {
    box-sizing: border-box;
}
```

```
body {
    background-color: whitesmoke;
    color: #333;
    font-family: 'Inter', sans-serif;
    margin: 0;
    padding: 40px 20px;
    display: flex;
    justify-content: center;
}

.container {
    width: 100%;
    max-width: 720px;
    background-color: white;
    padding: 35px 30px;
    border: 1px solid lightgray;
    border-radius: 12px;
    box-shadow: 0 4px 15px lightgray;
    text-align: center;
}

h2 {
    color: darkblue;
    margin-top: 0;
    margin-bottom: 8px;
    font-size: 24px;
}

.subtitle {
    font-size: 14px;
    color: #555;
    margin-bottom: 24px;
}

.table-wrapper {
    overflow-x: auto;
    margin-bottom: 20px;
}

.schedule-table {
    width: 100%;
    border-collapse: collapse;
    border: 2px solid #555;
    background-color: #ffffff;
}

.schedule-table th,
.schedule-table td {
    border: 1px solid #777;
    padding: 10px 12px;
    text-align: center;
    font-size: 14px;
}

.th-main {
    background-color: #e8ecf2;
    font-weight: 700;
    color: #1a237e;
}

.th-sub {
    background-color: #f1f4f8;
    font-weight: 600;
    color: #333;
}

.day-cell {
    background-color: #f7f9fc;
    font-weight: 600;
```

```
    cursor: pointer;
    transition: background-color 0.2s;
}

.day-cell:hover {
    background-color: #e3edf9;
}

.topic-cell {
    background-color: #ffffff;
    font-weight: 500;
    cursor: pointer;
    transition: background-color 0.2s;
}

.topic-cell:hover {
    background-color: #f0f4ff;
}

.slot-yellow {
    background-color: #fff9c4;
    cursor: pointer;
    transition: filter 0.2s;
}

.slot-yellow:hover {
    filter: brightness(0.92);
}

.slot-purple {
    background-color: #c5cae9;
    cursor: pointer;
    transition: filter 0.2s;
}

.slot-purple:hover {
    filter: brightness(0.92);
}

.slot-lightgreen {
    background-color: #c8e6c9;
    cursor: pointer;
    transition: filter 0.2s;
}

.slot-lightgreen:hover {
    filter: brightness(0.92);
}

.edit-panel {
    background-color: #f8f9fa;
    border: 1px solid #e0e0e0;
    border-radius: 8px;
    padding: 16px;
    margin-top: 20px;
    margin-bottom: 20px;
}

footer {
    text-align: center;
    margin-top: 30px;
    font-size: 14px;
    color: gray;
    border-top: 1px solid lightgray;
    padding-top: 15px;
}
```

Case Study 9.js

```
let cells = document.querySelectorAll("#scheduleTable tbody td");
let editInput = document.getElementById("editInput");
let saveBtn = document.getElementById("saveBtn");
let resetBtn = document.getElementById("resetBtn");
let selectedInfo = document.getElementById("selectedInfo");

let currentCellIndex = null;

// Assign an index to each cell and load saved modifications
cells.forEach(function(cell, index) {
    cell.setAttribute("data-cell-index", index);

    cell.addEventListener("click", function() {
        currentCellIndex = index;
        let day = this.getAttribute("data-day") || "N/A";
        let time = this.getAttribute("data-time") || "N/A";
        let text = this.innerText.trim();

        // Update edit input
        if (editInput) {
            editInput.value = text;
            editInput.focus();
        }

        if (selectedInfo) {
            selectedInfo.innerText = "Selected (" + day + ", " + time + "): " + text;
        }

        // Save current active selection to sessionStorage
        sessionStorage.setItem("lastSelectedCell", JSON.stringify({
            index: index,
            day: day,
            time: time,
            content: text
        }));

        // Show alert box on click event
        alert("Schedule Details:\nDay: " + day + "\nTime: " + time + "\nTopic/Session: " + text);
    });
});

// Save edited content to localStorage and sessionStorage
if (saveBtn) {
    saveBtn.addEventListener("click", function() {
        if (currentCellIndex === null) {
            alert("Please click on a table cell first to select what to edit.");
            return;
        }

        let newText = editInput.value.trim();
        if (newText === "") {
            alert("Please enter some text to save.");
            return;
        }

        let targetCell = cells[currentCellIndex];
        targetCell.innerText = newText;

        // Retrieve existing saved edits from localStorage
        let edits = JSON.parse(localStorage.getItem("scheduleEdits") || "{}");
        edits[currentCellIndex] = newText;

        // Save to localStorage (persists across page reloads)
        localStorage.setItem("scheduleEdits", JSON.stringify(edits));

        // Save last edit activity to sessionStorage
        sessionStorage.setItem("lastEditedTime", new Date().toLocaleTimeString());

        if (selectedInfo) {
```

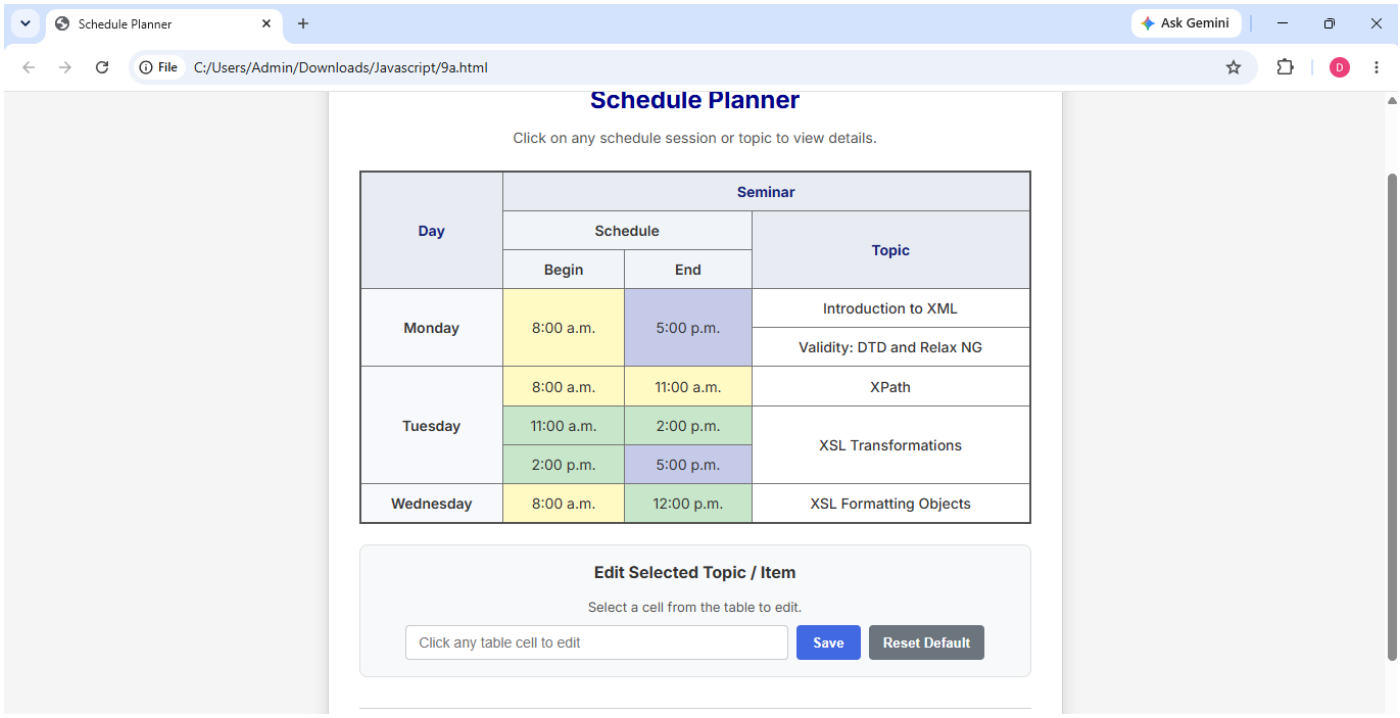
```
            selectedInfo.innerText = "Saved successfully: " + newText;
        }
        alert("Schedule updated and saved to storage!");
    });
}

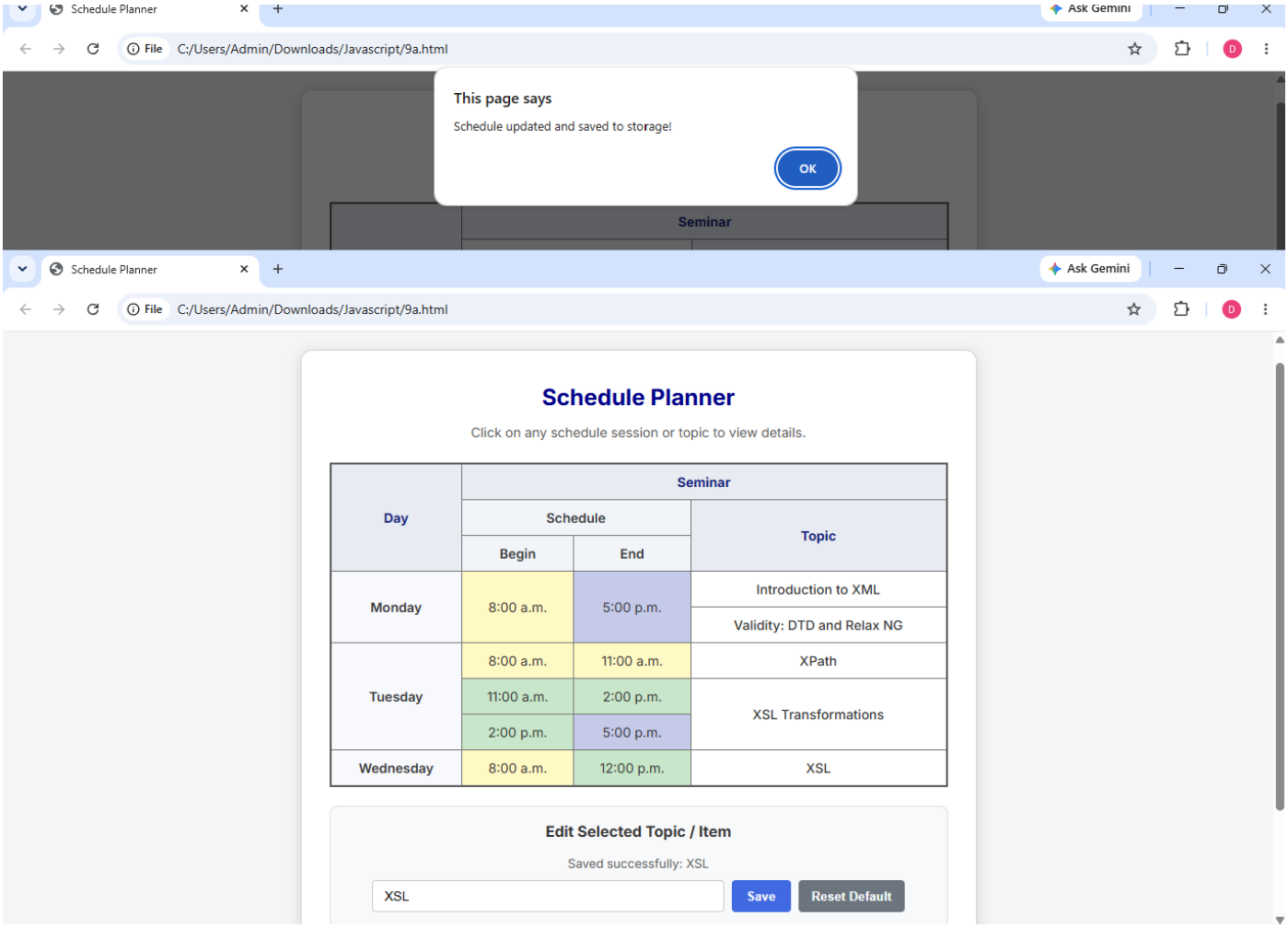
// Reset edits and clear localStorage
if (resetBtn) {
    resetBtn.addEventListener("click", function() {
        localStorage.removeItem("scheduleEdits");
        sessionStorage.removeItem("lastSelectedCell");
        sessionStorage.removeItem("lastEditedTime");
        location.reload();
    });
}

// Load saved edits from localStorage on page load
window.addEventListener("DOMContentLoaded", function() {
    let edits = JSON.parse(localStorage.getItem("scheduleEdits") || "{}");
    for (let index in edits) {
        if (cells[index]) {
            cells[index].innerText = edits[index];
        }
    }

    // Check if there was a previous session selection
    let sessionData = sessionStorage.getItem("lastSelectedCell");
    if (sessionData && selectedInfo) {
        let parsed = JSON.parse(sessionData);
        selectedInfo.innerText = "Resumed session cell (" + parsed.day + "): " + parsed.content;
    }
});
```

4. Output:


Schedule Planner
Ask Gemini
File C:/Users/Admin/Downloads/Javascript/9a.html
Schedule Planner
Click on any schedule session or topic to view details.
Day
Seminar
Schedule
Begin
End
Topic
Monday
8:00 a.m.
5:00 p.m.
Introduction to XML
Validity: DTD and Relax NG
Tuesday
8:00 a.m.
11:00 a.m.
XPath
11:00 a.m.
2:00 p.m.
XSL Transformations
2:00 p.m.
5:00 p.m.
Wednesday
8:00 a.m.
12:00 p.m.
XSL Formatting Objects
Edit Selected Topic / Item
Select a cell from the table to edit.
Click any table cell to edit
Save
Reset Default


Schedule Planner
Ask Gemini
File C:/Users/Admin/Downloads/Javascript/9a.html
This page says
Schedule updated and saved to storage!
OK
Seminar
Schedule Planner
File C:/Users/Admin/Downloads/Javascript/9a.html
Schedule Planner
Click on any schedule session or topic to view details.
Day
Seminar
Schedule
Begin
End
Topic
Monday
8:00 a.m.
5:00 p.m.
Introduction to XML
Validity: DTD and Relax NG
Tuesday
8:00 a.m.
11:00 a.m.
XPath
11:00 a.m.
2:00 p.m.
XSL Transformations
2:00 p.m.
5:00 p.m.
Wednesday
8:00 a.m.
12:00 p.m.
XSL
Edit Selected Topic / Item
Saved successfully: XSL
XSL
Save
Reset Default

5. Result/Conclusion:

An interactive schedule planner was built where clicking a table cell shows an alert box with the session details, and the selected schedule data is synced with both sessionStorage and localStorage so that it persists across page reloads. This case study reinforced table-based DOM events combined with persistent web storage in JavaScript.